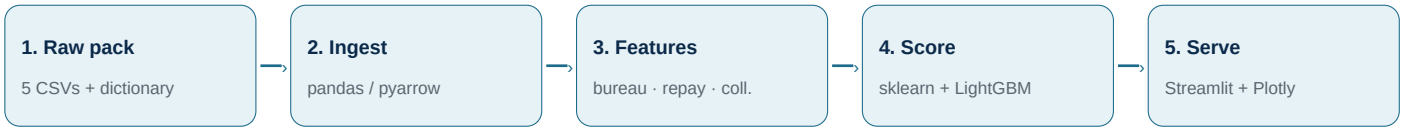
Architecture

Raw tables → clean panel → model & AI features → Streamlit (Community Cloud). One page.



Pipeline and tools

Grain: one row per loan account. loan_accounts LEFT JOIN customers on customer_id; LEFT JOIN bureau features rolled to customer; repayment and collections rolled to account_id.

- Ingest (pandas, pyarrow): map gender M/F→Male/Female; impute years_employed (by employment), dependents, residence years, interest rate (by product), bureau DPD missing→0. Drop only unusable target/amount. Deduplicate ids.
- Analytics: group-bys → data/processed/facts.json. This JSON is the only numeric payload the LLM may see.
- Model: sklearn OneHot+scaler+LogisticRegression (champion on test Gini) and LightGBM on bureau utilisation/DPD/write-offs, repayment DPD/missed EMI/paid ratio, collections counts. joblib in models/.
- Anomaly: IsolationForest on bureau/repay/collections mix plus peer z-scores (loan_type × region). Top 200 review queue.
- Dashboard: Streamlit + Plotly. Sidebar filters apply to every tab. Hosted on Streamlit Community Cloud (Python 3.11).

Ask the Portfolio — LangGraph

Graph: load facts → classify → retrieve evidence → generate → validate numbers → retry (max 2) or local fallback.

Where the LLM is called: generate node only, temperature 0, JSON response. Provider order: Groq openai/gpt-oss-20b (GROQ_API_KEY / Streamlit secrets) → Gemini → OpenAI. If no key: local retriever, still cited.

Sent to the LLM

Question + last few chat turns + slim facts.json (book totals and grouped default rates). Never account_id, customer_id, names, or raw rows.

Guardrails

Classify declines GDP/forecasts/PII/ACC ids. Validate rejects any % or $ not in evidence and falls back to local facts.

Anomaly detection — 200 accounts

sklearn IsolationForest + peer z on max DPD, write-off rate, missed EMIs, collections intensity. Each flag has a reason string. Confirm / Dismiss is stored for QA (session/local JSON). Not sent to the LLM.